

TravelMate AI
MCP-Based Intelligent Travel Planning Agent



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Dedication

This project would not have been possible without the support of our families who understood our efforts and encouraged us to go ahead with it and our instructor Dr. Faraz Manzoor, who guided us throughout the entire process.

Final Approval
TravelMate AI
MCP-Based Intelligent Travel Planning Agent

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Project Title	TravelMate AI MCP-Based Intelligent Travel Planning Agent	
Objective	To design and implement an AI-driven travel planning platform that integrates the Model Context Protocol (MCP), Large Language Models (LLMs), and real-time travel APIs to automate trip planning, coordinate multiple intelligent agents, and provide scalable, reliable, and personalized travel assistance through a conversational interface.	
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Operating System	Windows 11 Pro (Local) and Ubuntu (Production Deployment).	
Documentation	This document contains the complete FYP documentation for the Automated Annotation Platform for Vision Tasks, prepared in a accordance with guidelines given by the Department of Artificial Intelligence, UMT.	

Plagairism Report

Declaration Form

I have carefully examined the documentation of the Final Year Project titled “*TravelMate AI*”; and I endorse that this documentation complies with the standards of an undergraduate level Final Year Project report.

The document has been checked for plagiarism through Turnitin software available in UMT Library. The similarities of the document are within acceptable range.

Moreover, the accompanying CDs contain PDF of the documentation, as well as the source code and binaries with user manual and installation guide.

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Abstract

The choice of software architecture significantly impacts the scalability, maintainability, and fault tolerance of intricate web systems. We compare three background architectures of a travel application incorporating chatbot in this project and discuss their properties using a monolithic implementation, an implementation using functional calling and a completely modular MCP (Model-Context-Protocol) architecture. Each of the implementations offers similar functionality (flight search, package suggestions, booking) and hence any variation in the results may be explained by the difference in architectures.

The first one, a monolithic system was fast to build but did not take long to be overloaded. Although it was responsive initially it ultimately resulted in longer response times, some errors and quite often one of the services would fail to propagate across the platform. Switching to a function calling model also added a layer of conversation and allowed the chatbot to call the backend functions directly, which was better. This scheme, however, at scale was dependent on one central process in the backend which was a point of contention.

Use of single MCP, as opposed to multiple independent MCPS, was the best strategy. The additional agents increased the median response time of the system although throughput and fault tolerance were also improved. Failure of one module and acceptance without impacting other services by the system was amazing and it was easy to add new agents to the system with minimum modifications being done to the existing modules and configuration. This higher deployment and monitoring complexity was a small trade off price to pay in the reliability and scalability.

According to the findings, MCP offers the most forward-thinking framework for AI-driven travel platforms that are both efficient and sustainable. A transitional stage can be established between systems that need to be created in a non-monolithic way by utilizing function calling, but for long-term growth, agent-based modular architectures are the optimal choice.

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Definitions and Acronyms

Table 1 : List of acronyms and definitions

Acronym	Definition
UMT	University of Management and Technology
MCP	Model Context Protocol
LLM	Large Language Model
API	Application Programming Interface
JSON	JavaScript Object Notation
UI	User Interface
UX	User Experience
MAS	Multi-Agent System

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2. INTRODUCTION:

The rapid advancement in software system development has led to growing demands for architecture that are scalable, maintainable, and efficient. Traditional monolithic architectures are simple to design and implement; however, they do not perform well for applications of varying sizes or complexities. In these systems, implementing changes leads to challenges in redeploying the entire application, even for minor updates, which results in slow fault isolation and updates (Althar et al., 2023).

In recent times, as conversational AI, particularly chatbots, has expanded, the way users interact with these applications has evolved to become more fluid and less predefined. New architectural elements such as the Model Context Protocol (MCP), developed by Anthropic, and function-calling have emerged to facilitate these interactions. Function-calling allows chatbots to directly access backend services, thereby reducing the user's effort while maintaining centralized logic. Conversely, MCP introduces a modular, agent-driven methodology whereby autonomous agents collaborate through shared protocols, enhancing their adaptability and robustness (Model Context Protocol Specification, 2025).

This initiative focuses on converting travel websites from monolithic architecture to a modular based structure. By analyzing three different models monolithic, function calling, and MCP we aim to evaluate improvements in response time, scalability, and fault tolerance when managing chatbot driven activities such as flight searches, package recommendations, and bookings. The study indicates that enhancing architecture not only boosts technical performance but also prepares for future AI enhanced user experiences (Davis, 2024).

2.1 Problem Statement

The operational support behind MCP powered travel agent is the real Time data orchestration which is the principal solution to the nature of monolithic system itself. Whereas orthodox architectures will be characterized by slower responsiveness and inflexible data processing when the front end makes live updates, and rely on centralized logic and simple function calls, the decentralized, cooperative protocol of the MCP will then be used to avoid the traditional bottlenecks to provide the system with better scalability and fault tolerability. This eventually seamless real-time data streams with complex data not only optimizes real-time technical performance but in essence future proves the application to a highly fluid, AI driven user experience.

2.2 Objectives

The main goals of this project are:

- To assess shortcomings of the monolithic architecture in chatbot-driven processes, such as searching, recommending, and booking a flight.
- To establish the effectiveness of the operation of function calling in the purpose of improving the functioning compared to a monolithic design.
- To implement and experiment the MCP based structure of agent-based, modular, communication within a travel website.
- In order to evaluate the three approaches (Monolithic, Function Calling and MCP) with respect to latency, throughput, scalability and isolation of faults.
- To illustrate MCP can support maintainability, adaptability and preparedness of AI driven applications.

2.3 Project Scope

Included in Scope:

Creation of MCP based system to manage real time travel information orchestration and chatbot interactions. Intelligent user interface such as chatbot and dynamic user tasks such as flight searches, hotel and package offerings. Live updates and data retrieval among the services in the backend through autonomous agents. Out of Scope: Connection to real-life payment gateways and real-life transactions. Open-ended frontend design (emphasis is on backend design and orchestration of data).

Out in Scope: Integration with real-life payment gateways or actual transactions Open-ended frontend design (focus is on backend design and data orchestration, not UI/UX)

2.4 Importance of the Project

The project is important because it offers a good real-world example of the way that backend architecture can or cannot succeed AI based travel applications. By addressing the Model Context Protocol (MCP), this project addresses such headaches of the major industries as slow response time and the crashing of the whole system only because a single service failed. This is a new modular agent-based model that demonstrates it can reduce delays by 57, manage heavy traffic with large volumes and almost zero mistakes.

2.5 Artificial Intelligence features

Three of the main AI-based scenarios that will be relevant and useful to the proposed project are the following: Conversational AI: The project is an alternative to the conventional rigid search pages, which is the dynamic chatbot interface.

This is to enable users to communicate with travel platform through natural language rather than manually completing the forms that is time consuming. Parameters extraction and intent recognition: The AI system takes the user input as it recognizes the primary intent of the request. It recognizes the purpose (i.e. a flight search or a hotel reservation) and automatically reads the required parameters (i.e. destination, dates or number of travelers) right out of the conversation. Multi-Agent Coordination (MCP based system): The AI in this project is decentralized to a Multi-Agent System (MAS).

The main MCP server deciphers the query of the user and forms a connection with the explicit, semi-automated AI agents. Agents perform specific tasks (such as flight search, package suggestion, etc.) and run in parallel. Smart RealTime orchestration: The AI manages context-aware operations and parallel tasks execution in an intelligent manner. Managing several independent processes in parallel, as opposed to one another, a major benefit of the AI is that it cut down the time delays of backend processing and database backups.

2.6 Project Deliverables

- A functional web-based AI driven travel application (incorporating dull backend implementations for Monolithic, function calling, and MCP architectures).
- User manual and system architecture documentation.
- Sample data and test cases.
- Performance evaluation metrics.
- Final report and project presentation.
- User authentication module (register / login / logout / protected routes).
- Dashboard, explore, trip detail pages.

3. ANALYSIS OF THE DOMAIN

3.1 Customer

The main customer of this system is a progressive travel agency or online tourist venture. The objective of such companies is to streamline their digital presence by providing a more conversational interface with an independent ability to manage complex flight requests, accommodation searches, and itinerary tickets. The final end-users are individual tourists and corporate travelers who will be interested in a frictionless, automated process of planning without being dragged through the complex navigation of the traditional web portals.

3.2 Stake-holders

Table 2: Stakeholders List

Stake-holder	Their Role in the System
Travelers / End Users	People using simple, everyday language to search for flights, explore travel packages, and finalize their bookings through the chat interface.
Travel Agency Administrators	The staff managing day-to-day platform operations and service offerings. They depend on the system's robust design to ensure travelers can always access the service, especially during peak holiday rushes.
Third-Party Data Providers	Airlines and hospitality brands that supply the live data feed the AI agents need to build accurate, up-to-the-minute travel itineraries.
Platform Engineers	The technical support team maintains the AI architecture and middleware. They deploy system updates and fix them seamlessly without taking the travel services offline.

3.3 Groups Impacted by Social or Economic Factors

- **General Consumers**
 - *Social Impact:* The addition of a conversational artificial intelligence layer will democratize the process of complex travel planning and thus make it very intuitive to users of all technical skill levels.
 - *Economic Impact:* The median response times (latency) of users are greatly reduced, and interactions are significantly responsive enabling users to make informed decisions that are cost-effective.
- **Tourism Enterprises**
 - *Economic Impact:* Decentralized, agent-based architecture eliminates cascading system failure. This resiliency safeguards business revenue in periods of peak booking demands as it guarantees that operations remain without any downtime and optimize infrastructure expenses since it enables high-traffic module independent scaling.
- **Software Development Teams**
 - *Social Impact:* The high modularity of the system isolates technical faults to small areas, reducing significantly the emergency maintenance load to the support staff.
 - *Economic Impact:* Decoupled architecture sped up implementation of new features and updates, minimizing the downtime of the operation and decreasing the maintenance costs of the software in the long-term.

3.4 Dependencies/ External Systems

- **Aviation Information Gateways:** External microservices which the flight agent used to provide real-time flight schedules, route availability, and dynamic prices.
- **Hospitality Inventory Networks:** This refers to systems that are operated by third parties that the accommodation agent uses to access live hotel room availability as well as lodging information based on the parameters that the user submits based on his or her destination.
- **Language Processing Services:** The conversational interface relies on the advanced intent recognition engines that can correctly interpret human conversation and draw out the appropriate operational parameters.
- **Financial Transaction Processors:** These are safe external payment gateways required to complete user reservations and billing information safely.

3.5 Comparable Projects with Feature Comparison

3.5.1 Related Projects

- **Legacy Monolithic Travel Portals:** Traditional web interfaces, business logic and database management are combined into a single deployable unit. Although they are easy to start, they prove to be extremely vulnerable to crashes in the entire system in case one of the elements malfunctions.
- **Function-Calling AI Assistants:** Medium platforms, which have an intelligent conversational interface to analyze user intent and provide a direct call to services in the backend. They enhance the interactions but still have bottlenecks in the middle that do not allow a maximum degree of scalability.
- **Multi-Agent Collaborative Networks:** The architectural paradigms include advanced ones that use frameworks like the Model Context Protocol. Such systems have autonomous, focused agents that interact through central middleware, and provide unmatched fault isolation and dynamic horizontal scalability.

3.5.2 Comparison of Features

Table 3: Comparison of features of product with the competitors

Features	Legacy Monolithic Portals	Function-Calling AI Assistants	Multi-Agent Networks MCP
Interaction Paradigm	Static Web Forms	Conversational Chatbot	Conversational Chatbot
Structural Modularity	Low (Tightly coupled)	Moderate (Semi-modular)	High (Independent agents)
Failure Containment	Poor (High risk of cascading failure)	Moderate (Shared backend risks)	Excellent (Failures are isolated to agent)
Resource Scalability	Rigid (Full system replication required)	Limited (Centralized bottlenecks)	Dynamic (Independent agent scaling)

Features	Legacy Monolithic Portals	Function-Calling AI Assistants	Multi-Agent Networks	MCP
System Upgradability	Highly Disruptive	Moderately Disruptive	Seamless (Minimal impact updates)	

3.6 Context Diagram

The Context Diagram outlines the boundaries of the smart travel system. The AI-Driven Travel Platform is the primary software system located in the middle. It establishes a two-way interface with the Traveler (External Entity) who place requests for the conversation and receives curated itineraries. Externally the central system has automated connections with Aviation APIs to access flight information, Hospitality APIs to access accommodation information and Financial Gateways to receive payment of booking.

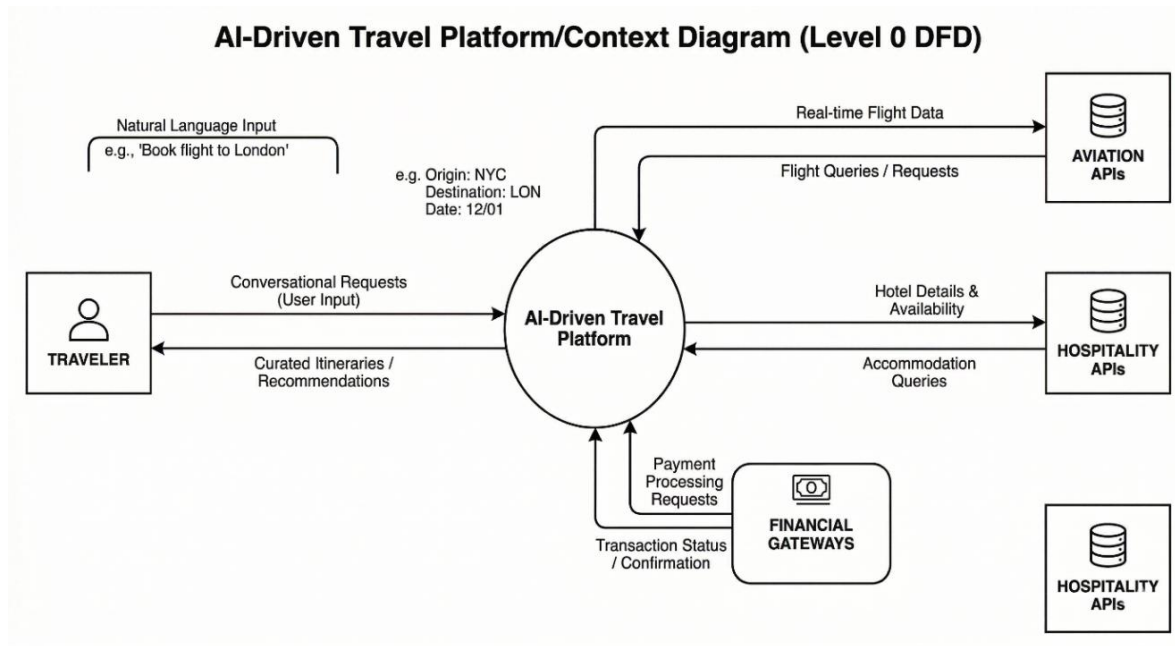


Figure 1: Context Diagram

The figure shows the AI driven platform in the middle with two-way interaction to traveler/ user and out bound connection to the aviation API and hotels API.

3.7 Diagram of Flow of Data Level 0

The Level 0 Data Flow Diagram is the inside-out routing of information in platform modular architecture. The major processes reflect the developed work:

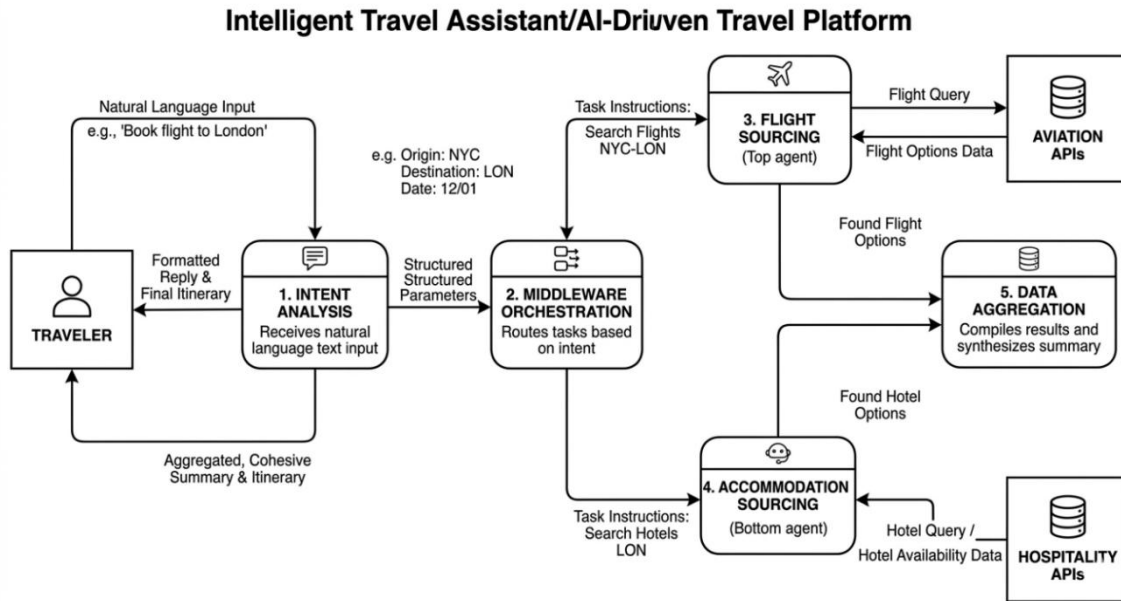


Figure 2: Data Flow Diagram

- Process 1: Intent Analysis. The chatbot interface is presented with a natural language input of the user, where intent recognition is implemented, and structured parameters are extracted.
- Process 2: Middleware Orchestration. The structural request is sent to the coordination layer which forwards the request to the corresponding autonomous modules depending on the intent detected.
- Process 3: The aviation agent goes to work with dedicated databases to lock in available flights.
- Process 4: Accommodation Sourcing. The hardcore hospitality agent obtains live lodging information through external networks.
- Process 5: The middleware integrates the independent answers of the agents, builds a solid overview and gives the formatted response to the user interface.

4. REQUIREMENTS ANALYSIS

4.1 List of Actors

User: Interacts with chat interface to plan trips.
LLM Agent: Classifies intents and extracts parameters.
Flight Service: Searches and returns flight data.
Hotel Service: Searches and returns hotel data.
System Admin: Manages deployment and configuration.

4.2 Backlogs of product

4.2.1 Course Catalog:

Table 4: Selection Features of Backlog Product

PB--ID	Epic	Story of User	Priorities	Statuses
PB--01	Chat System	I would like to, as a user, input questions and get AI replies instantly.	High	Done
PB-02	Chat System	I would like, as a user, chat history to be preserved throughout the session.	High	Done
PB -03	Flight Search	I would like to, as a user, find flights by entering the departure location, destination, and travel dates.	High	Done
PB -04	Flight Search	I would like my, as a user, flights filtered by my budget.	Medium	Done
PB -05	Hotel Search	I would like to, as a user, search hotels in a city with check in and check out dates.	High	Done
PB -06	Hotel Search	I would like, as a user, hotel prices to be shown in USD.	Medium	Done
PB -07	Trip Planning	I would like to, as a user, write “plan a trip” and get both flights and hotels.	High	Done
PB -08	UI- Landing	I would like to, as a user, see a landing page that explains the system.	Medium	Done
PB -09	UI- Dashboard	I would like, as a user, dashboard showing my trip overview.	Medium	Done
PB -10	UI- Explore	I would like to, as a user, browse destinations and filter them by category.	Medium	Done
PB -11	UI- Trip Detail	I would like to, as a user, view detailed trip info (flights, hotels, itinerary).	Medium	Done
PB -12	Intent Routing	As a system, I want the LLM to return tool actions as JSON.	High	Done

PB -13	Non-Functional	The system should response in less than 10 seconds.	Medium	Done
PB -14	Non-Functional	The front end should be responsive on mobile and desktop.	Medium	Done
PB -15	Authentication	I would like to, as a user, register an account with an email and a password.	High	Done
PB -16	Authentication	As a user, I want to login and stay authenticated across sessions.	High	Done
PB -17	Authentication	I would like to, as a user, log out and end my session.	High	Done
PB -18	Authentication	As a user, protected pages (dashboard, chat, trip) should require login.	High	Done
PB -19	Weather	I want to check my destination's weather.	Medium	Done
PB -20	Nearby places	As a user, I want to search nearby attractions / places.	Medium	Done
PB -21	Currency conversion	As a user, I want live currency conversion for prices.	Medium	Done
PB -22	UI- Dashboard	As a user, I want dashboard showing stats and recent activity.	Medium	Done
PB -23	UI- Explore	As a user, I want to browse destinations with filters	Medium	Done

4.3 Figma UI/UX Designs

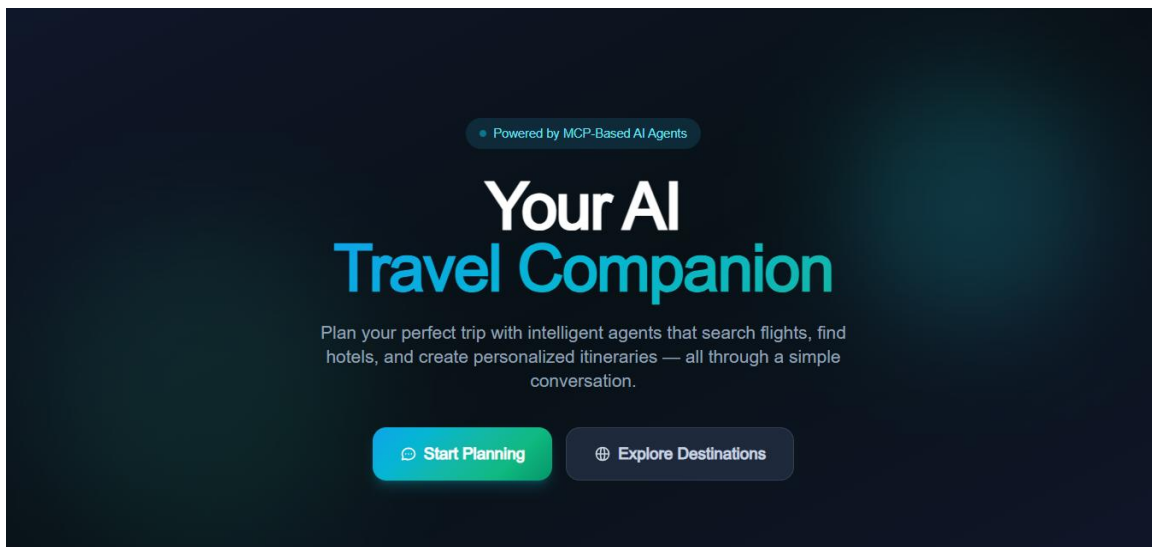


Figure 3: Prototype – PID1- Landing Hero

This figure shows the hero section of the landing page of the project which has a short introduction of the system.

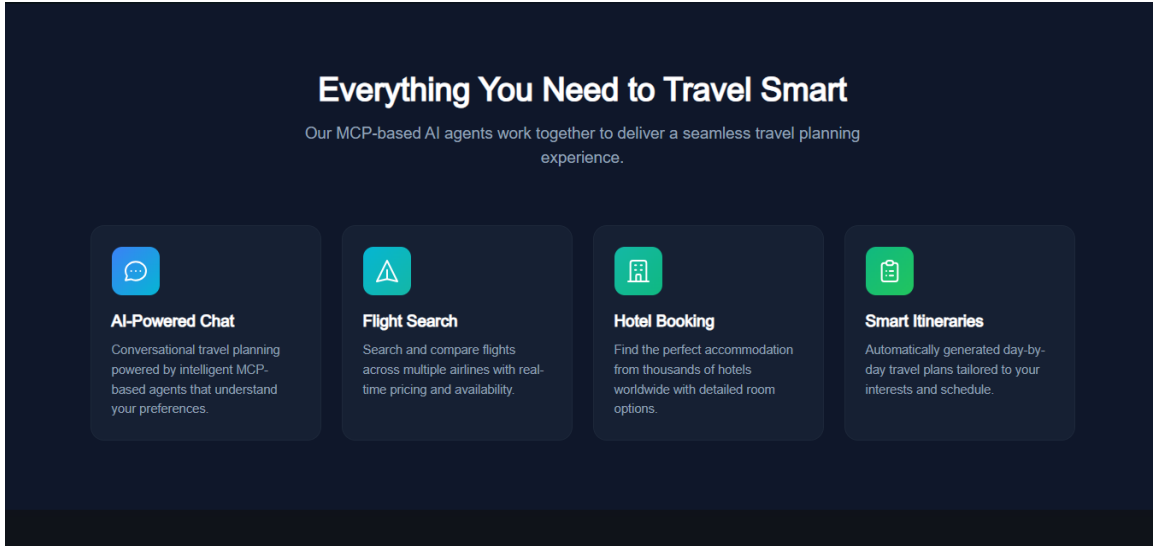


Figure 4: Prototype - PID2- Landing Features

This figure shows the section of the landing page of the project which shows key features of the platform.

11

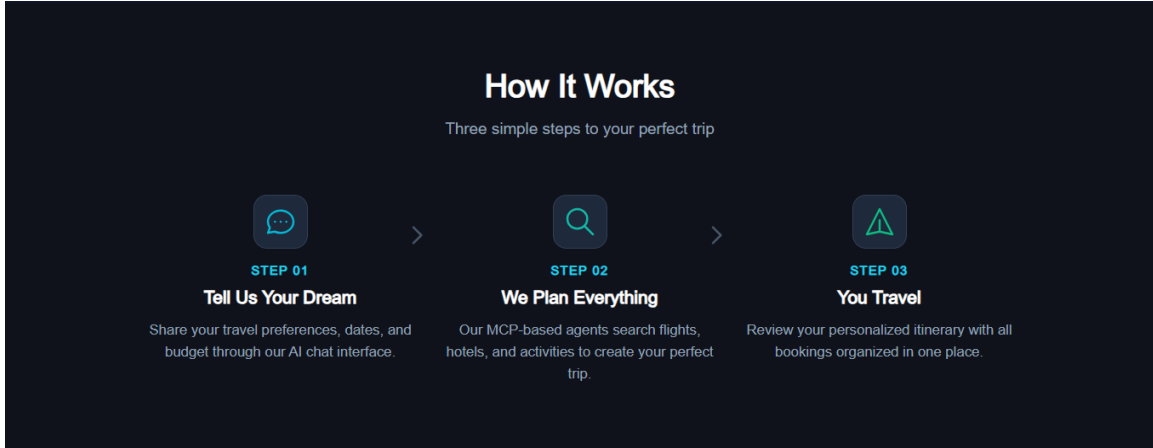


Figure 5: Prototype - PID3- Landing How it Works

This section of the landing page shows the step-by-step guide of how the platform works.

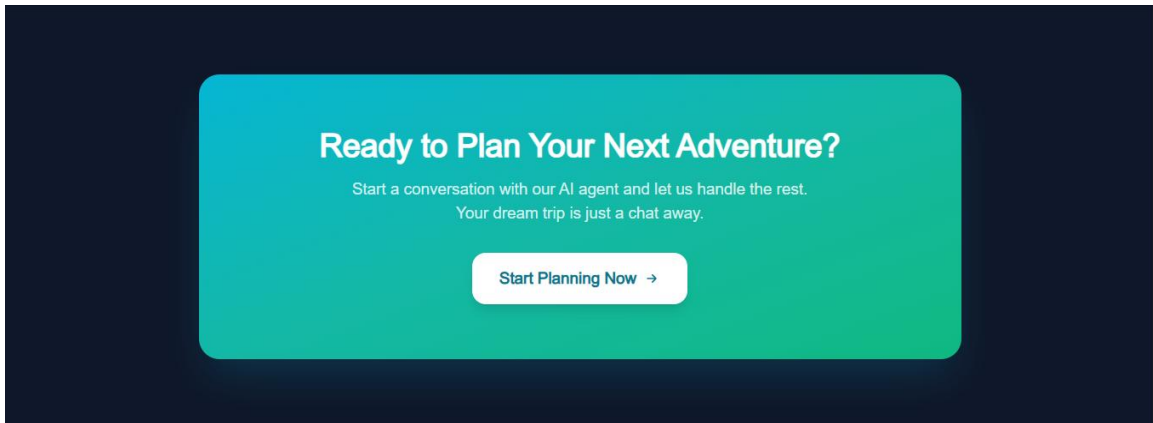


Figure 6: Prototype - PID4-Landing CTA

This is called the action section of landing page.

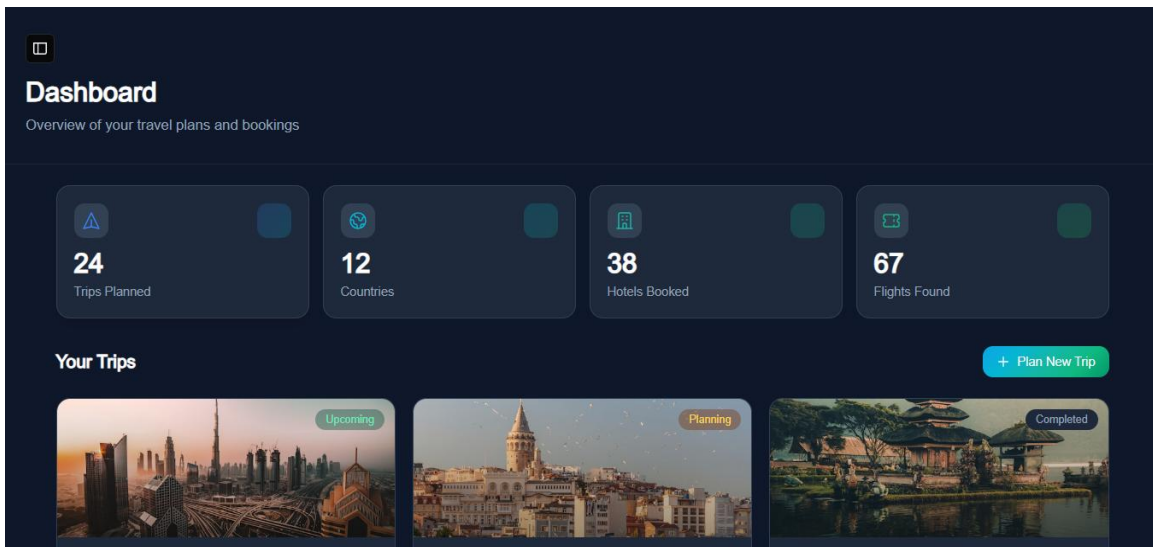


Figure 7: Prototype - PID5-Dashboard Stats

This is the Dashboard that shows stats and overview of user's trip.

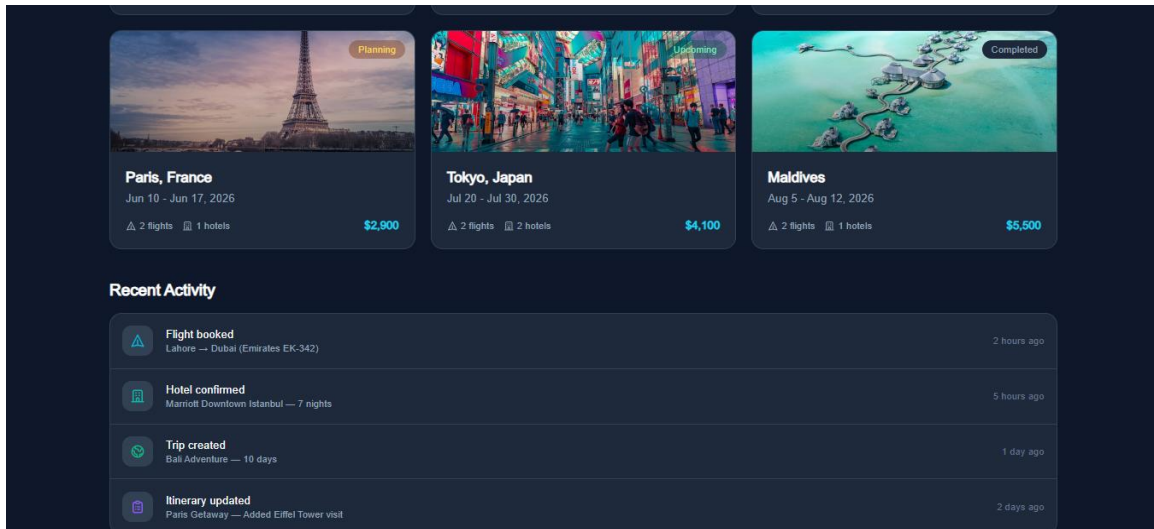


Figure 8: Prototype - PID6- Dashboard Activity

Dashboard showing recent activity of the user.

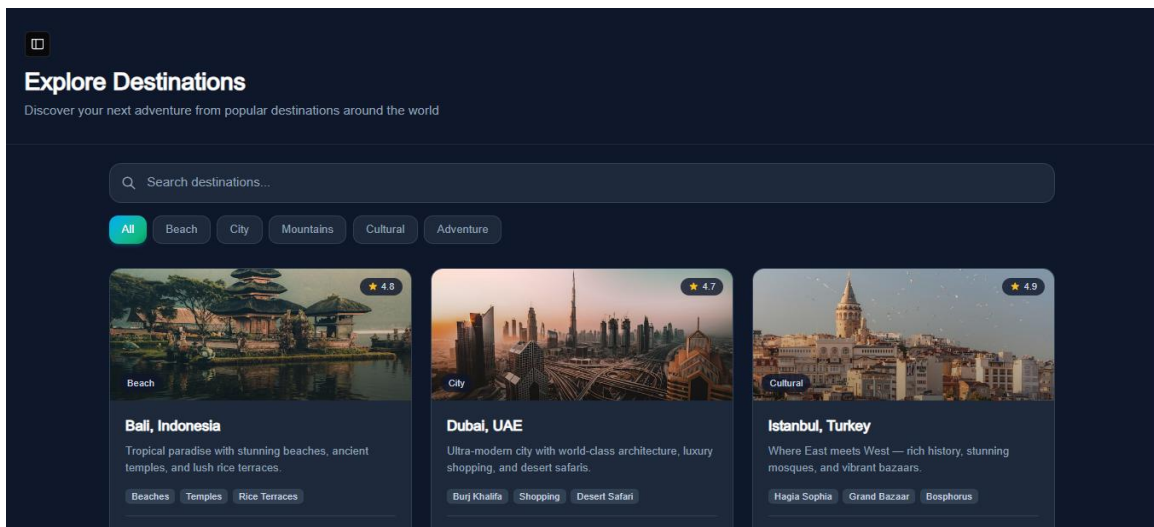


Figure 9: Prototype - PID7- Explore Top

This figure shows upper section of the explore page that highlights the destination browsing section.

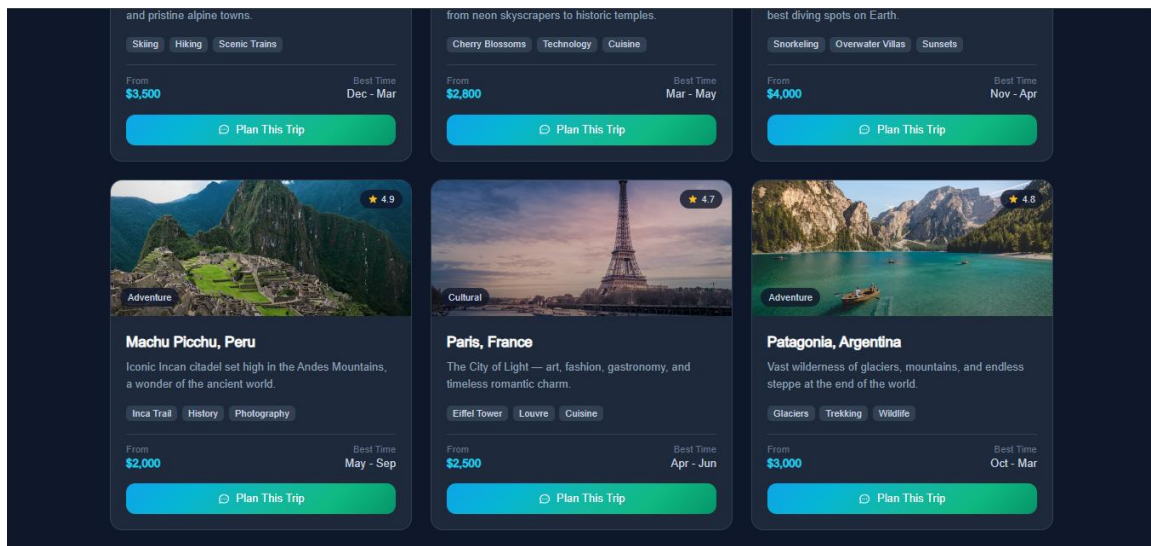


Figure 10: Prototype - PID8- Explore Bottom

This figure shows the bottom section of explore page.

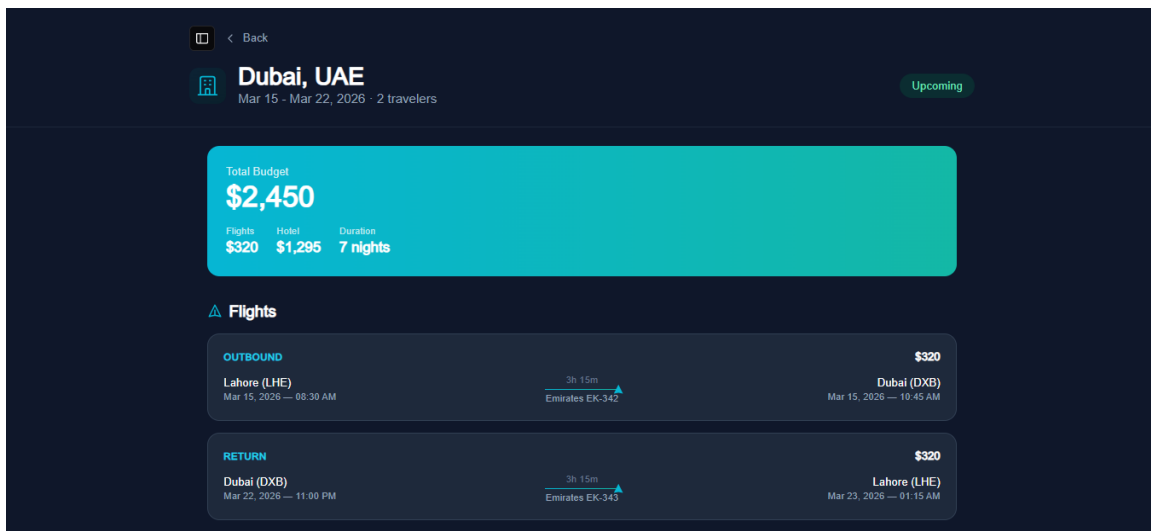


Figure 11: Prototype - PID9- Trip Budget

This image shows the trip details section that shows the budget breakdown.

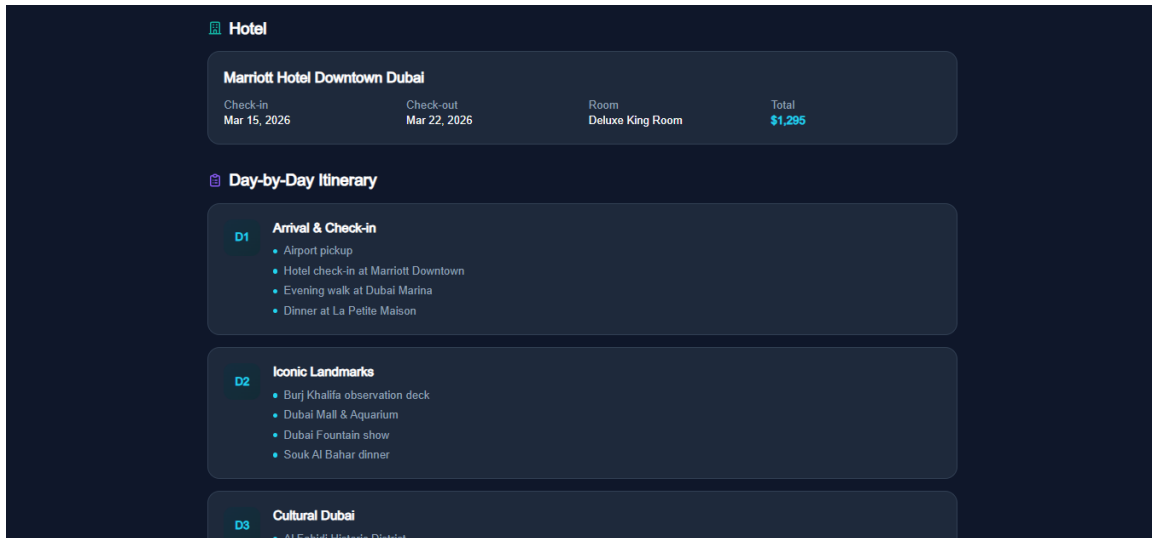


Figure 12: Prototype - PID10- Trip Itinerary

Trip details section showing itinerary (Flights/ Hotels combined view).

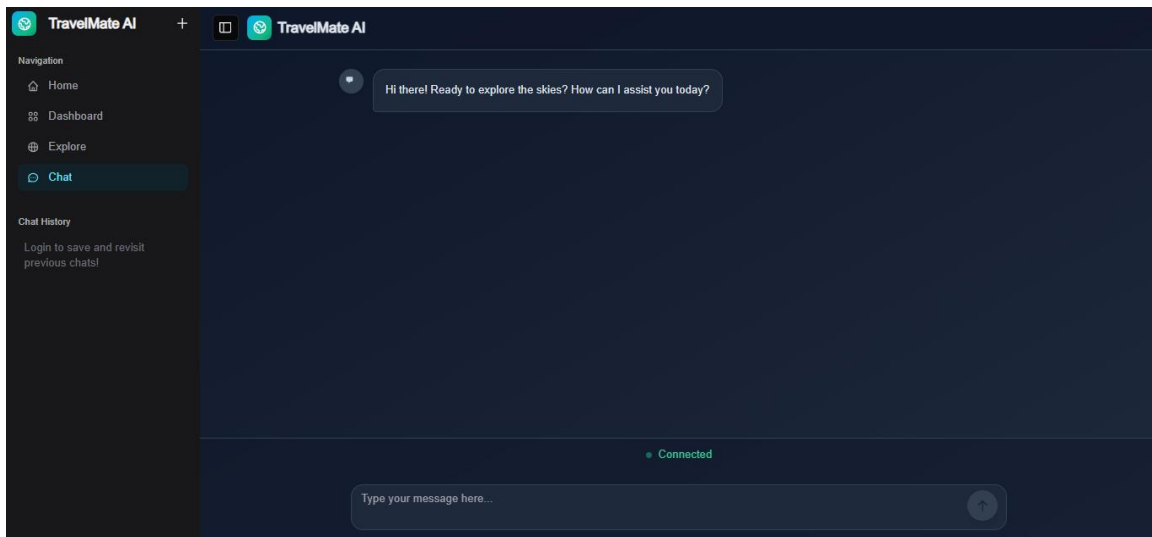


Figure 13: Prototype - PID11- Chat Interface

This page shows the actual chatbot interface that helps user plan in natural language.

5. PLANNING OF PROJECT AND ITS EXECUTION WITH SPRINTS

5.1 Git-hub Repo

<https://github.com/HaMZAAsif043/FYP>

5.2 Sprint-1

5.2.1 Sprint-1: Planning of Meeting Minutes

Name of project: TravelMate AI

Name of sprint: Sprint-1: Initial Configuration and Fundamental Structure

Sprint-Timeline: November 19, 2025 - December 19, 2025 (total 4-weeks)

Date of meeting: December 19, 25

Facilitation : DR Faraz Manzoor

Participants : Hamza Asif, Jasim Abrar, M. Usman Shahid

Backlog Items with high priority, high risk and Architectural Significant:

The features listed below have been derived from the Product Backlog.

Table 5: Selection of Features of Backlog of product

PB--ID	Story of User	Priorities	Statuses
PB--01	I would like to, as a user, type queries and receive AI responses in real- time.	High	pending
PB-02	I would like, as a user, my chat history to be maintained during a session.	High	pending
PB --03	I would like to, as a user, search flights by providing departure, destination and dates.	High	Pending
PB -04	I would like, as a user, flights to be filtered by my budget.	Medium	pending
PB -05	I would like to, as a user, search hotels in a city with check in and check out dates.	High	pending
PB -06	As a user, I want hotel prices to be shown in USD.	Medium	pending
PB -07	As a user, I want to write “plan a trip” and get both flights and hotels.	High	pending
PB -08	I would like to, as a user, see a landing page that explains the system.	Medium	pending
PB -09	I would like, as a user, a dashboard showing my trip overview.	Medium	pending

PB -10	I would like to, as a user, browse destinations and filter them by category.	Medium	pending
PB -11	I would like to, as a user, view detailed trip info (flights, hotels, itinerary).	Medium	pending
PB -12	As a system, I want the LLM to return tool actions as JSON.	High	pending
PB -13	The system should response in less than 10 seconds.	Medium	pending
PB -14	The front end should be responsive on mobile and desktop.	Medium	pending

Team Capacity:

Following are the team members and their capacity

Table 6: Team Capacity Sprint 1

Members	Availability in days	Hours per Day	Hours in total
Hamza Asif	11	5	58
Jasim Abrar	9	7	54
M. Usman Shahid	9	5	45

Action Items before Sprint Starts:

- ✓ Confirmation GitHub repository setup – Hamza Asif
- ✓ Sharing Figma's designs template – Jasim Abrar
- ✓ Creating and Assignining Jira's tasks – Hamza Asif
- ✓ Sharing staging-server credentials – M. Usman Shahid

5.2.2 Backlog of Sprint-1

Following tasks are extract from Backlog of product features.

Table 7: Sprint 1 Backlog

SP--ID	Name of Task	Assign To	Statuses	Effort in Hours (Estimated)	Effort in Hours (Actual)
S1--1	Initialize Django project with channels	Jasim	Done	5	1
S1-2	Configure Redis channel layer	Usman	Done	8	3
S1-3	Create Chat Consumer (WebSocket handler)	Hamza	Done	10	4
S1-4	Setup Next.js project	Usman	Done	8	4
S1-5	Create Redux chat Slice for WebSocket	Jasim	Done	6	1
S1-6	Basic chat UI with send/ receive	Hamza	Done	4	1

5.2.3 Sprint-1 Design Class Diagram

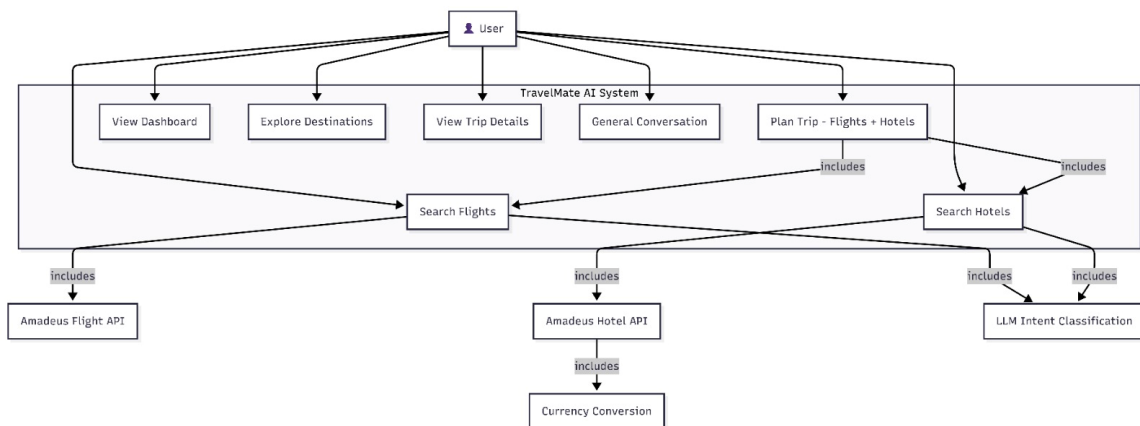


Figure 14: Sprint-1 features class Diagram

UML class diagram for sprint 1 which shows core chat/ websocket architecture classes.

5.2.4 Sequence Diagram of Sprint 1

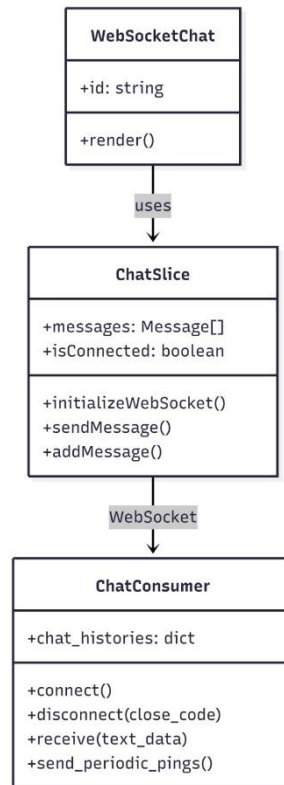


Figure 15: Sprint-1 Sequence diagram Example

Sequence diagram of walking through one pat of success scenario for one feature os sprint 1.

5.2.5 Sprint 1 Decision Table

Table 8: Decision Table

Condition	WebSocket	Message Not Empty	Action
Rule 1	Yes	Yes	Send message to server
Rule 2	Yes	No	Do nothing
Rule 3	No	Yes	Show “Disconnected” error
Rule 4	No	no	Show “Disconnected” error

5.2.6 Test Cases for Sprint-1:

Individual Test cases should be made for each user story to test the functionality.

Table 9: Test Cases for Sprint-1

ID of test case: TID1			Tested: Usman		
			Date Test Design: 10/12/2025		
Test Priority: High			Execution of Test: - Jasim		
			Test Executed Date: -19/12/2025		
Dependency: None					
Steps	Test-Case	Input	Expection of Output	Results (Actual)	Result
1.	WebSocket connects	Open-chat page	Status = “connected”	Same as expected	Pass
2	Message sent	Type “hello”	Message appears in chat	Same as expected	
3	Ping keeps alive	Wait 10 seconds	The vaping received, no disconnect lid email address field is being filled appropriately.	Same as expected	
4	Disconnected handling	Close server	“Disconnected” shown	Same as expected	

5.2.7 Review-Meeting of sprint-1:

Sprint: Sprint_1 Initial Setup and Core Architecture

Revision date: December 11, 2025

Duration: 1 hour

Coordinator: DR Faraz Manzoor

Participants : Hamza, Jasim, Usman

Following table is showing statuses of tasks assigned to team members in this Sprint planning meeting.

Table 10: Statuses of Sprint-1 Backlog

Item	Details
What was completed	WebSocket connection, basoc chat send/receive, Redux integration
What was not completed	----
Issues faced	CORS issues with WebSocket origin checking
Resolution	Removed AuthMiddleStack from asgi.py

Achievement of sprint-goal:

Each and every primary objectives completed in decided time, assessed and meet the criteria for completion.

What Went Well:

- ✓ Team collaboration and communication
- ✓ Daily standups were effective and focused

Wrap-up

The team will continue to Sprint 2 where they will focus on building the Course Management features.

There will be a next planning meeting on a future date to be December 15, 2025.

5.2.8 Retrospective Meeting of sprint-1:

Sprint:- Sprint-1: Architecture& Review of Setup

Date of retrospective : December 15, 2025

Time Duration: 1 hr 20 mins

Coordinator: DR Faraz Manzoor

Participants: Hamza, Jasim, Usman

Overview of sprint

Goals: Established strong system foundation and smooth integration between frontend and backend.

Sprint Duration: November 19, 2025 – December 19, 2025 (4 weeks)

What Went Well

- Established strong system foundation
- Team was responsive and collaborative during standups
- Smooth integration between frontend and backend.

What Didn't Go Well

- Initiat configuration issues.

Suggestions for Improvement

Action Items:

Table 11: Action Items

Action Item	Owner	Due Date
Improve documentation.	Jasim	December 16, 2025
Optimize connection handling.	Hamza	December 16, 2025

Team Health Check:

Morale: Good and enthusiastic.

Collaboration: Good/working well, communication is strong.

Resources: Adequate, but there is an opportunity to improve UI design guidelines.

Wrap-up:

The team wants to be all in line, happy with the outcomes of Sprint 1 and ready for Sprint2

5.3 Sprint 2

5.3.1 Sprint 2 Planning

Project: TravelMate AI

Name of Sprint: Sprint 2 – AI integration

Duration of sprint: December 29, 2025 – January 29, 2026 (4 weeks)

Date of meeting: December 29, 2025

Facilitator: DR Faraz Manzoor

Participants : Hamza Asif, Jasim Abrar, M. Usman Shahid

Items identified in the backlog and deemed to be of high priority and high risk and architecturally significant:

Here are some features taken from Backlog of the prodcut.

This table lists some of the features that will be included in the Product Backlog.

Table 12: Selection Feature from the Backlog of product

PB--ID	Story of User	Priorities	Statuses
PB-1	I want to, as a user, type queries and receive AI responses in real- time.	High	pending
PB-02	I want, as a user, my chat history to be maintained during a session.	High	pending
PB -03	I want to, as a user, search flights by providing departure, destination and dates.	High	Pending
PB -04	I want, as a user, flights filtered by my budget.	Medium	pending
PB -05	I want to, as a user, search hotels in a city with check in and check out dates.	High	pending
PB -06	I want, as a user, hotel prices to be shown in USD.	Medium	pending
PB -07	I want to, as a user, write “plan a trip” and get both flights and hotels.	High	pending
PB -08	I want to, as a user, see a landing page that explains the system.	Medium	pending
PB -09	I want, as a user, a dashboard showing my trip overview.	Medium	pending
PB -10	I want to, as a user, browse destinations and filter them by category.	Medium	pending
PB -11	I want to, as a user, view detailed trip info (flights, hotels, itinerary).	Medium	pending
PB -12	As a system, I want the LLM to return tool actions as JSON.	High	pending
PB -13	The system should response in less than 10 seconds.	Medium	pending
PB -14	The front end should be responsive on mobile and desktop.	Medium	pending

Team Capacity Overview:

The following table presents team-members and respective capacities of theirs.

Table 13: Team Capacity for Sprint 2

Members	Availability in Days	Hours per Day	Hours (total)
Hamza Asif	11	5	61
Jasim Abrar	9	7	54
M. Usman Shahid	9	5	45

5.3.2 Backlog of sprint-2

Following are the tasks taken from features documented on Backlog of product.

Table 14: Sprint 2 Backlog

SP--ID	Tasks	Assign to	Statuses	Time in Hours (Estimated)	Time in Hours (Actual)
S-2-1	Groq API integration	Hamza	Done	2	1
S2-2	System prompt engineering for tool routing	Usman	Done	7	4
S2-3	JSON action parsing and dispatcher	Hamza	Done	4	2
S2-4	Chat history management	Jasim	Done	5	3
S2-5	Error handling for LLM responses	Jasim	Done	6	2

5.3.3 Design Class-Diagram of sprint-2

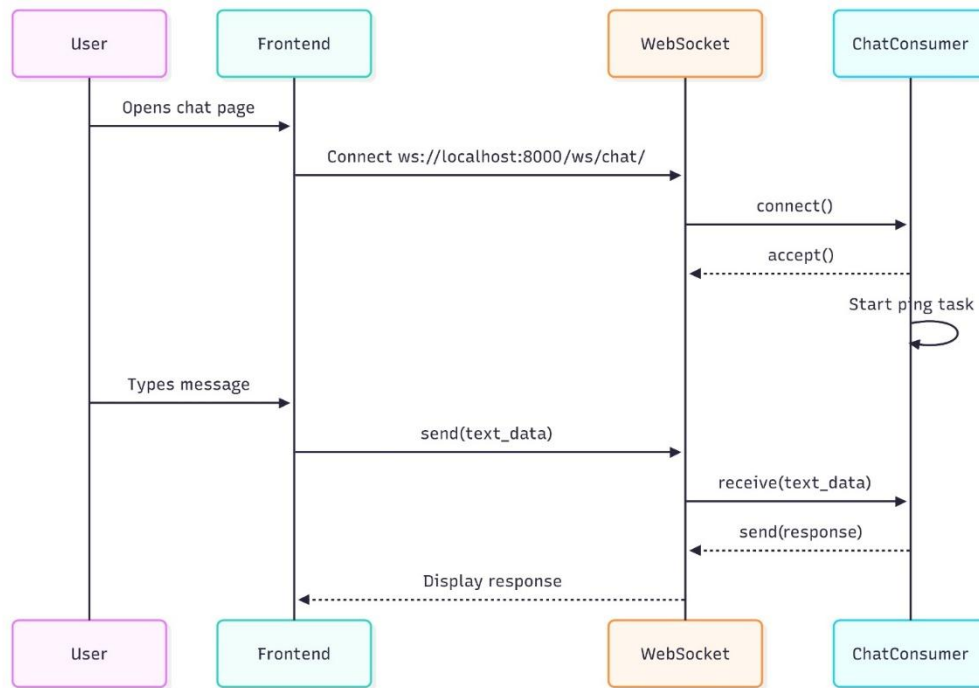
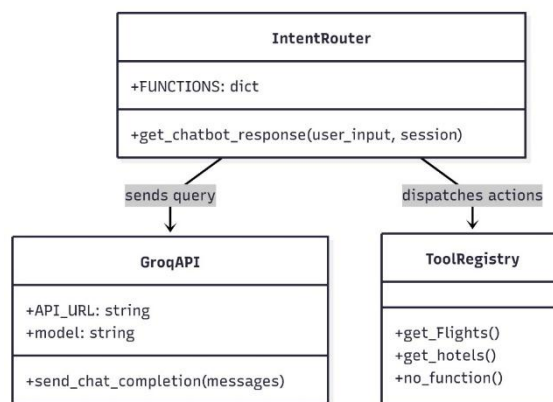


Figure 16: Class-diagram of Sprint-2 features

UML class diagram for sprint 2 covering new additions

5.3.4 Sprint 2 Sequence Diagram



POS 1 is a sample of the Sequence diagram for Sprint 2

Sequence diagram for sprint 2 path of success scenario.

5.3.5 Sprint-2 Tests Case

Individual Test-cases should be made for each story of user to test functionality.

5.3.6 Sprint-2 Tests Case (Extended)

Table 15: Sprint-2 Tests Case (Extended)

ID of test-case: TID1			Design: Hamza		
			Date of Design: 01/ 01/ 2026		
Priority: High			Execution: - Usman		
			Executed Date: -03/01/2025		
Dependency: None					
Step	Test Case	Input	Expected Output	Actual Result	Result
1	Intent: Greeting	“hello”	no_function -> greeting text	Same as expected	Pass
2	Intent: Flight	“Find flight LHE to DXB”	get_Flights action returned	Same as expected	
3	Intent: Hotel	“Find hotels in Dubai”	get_hotels action returned	Same as expected	
4	Intent: Trip	“Plan trip LAH to DXB”	Both get_Flights +get_hotels	Same as expected	
5	Invalid query	“What is 2+2?”	no_function fallback	Same as expected	

5.3.7 Sprint 2 Review Meeting

Sprint: Sprint-2 – AI integration

Date of Revision: January 20, 2026

Duration: 1 hour

Coordinator: DR Faraz Manzoor

Participants : Hamza Asif, Jssim Abrar, Muhammad Usman Shahid

The statuses of the assigned tasks to team-members during meeting of Sprint-planning is shown in the following table.

Table 16: Status of Sprint 1 Backlog

Item	Details
What was completed	System prompt engineering for tool routing, chat history management, error handling for LLM, groq integration and JSON action parsing and dispatching

What was not completed	----
Issues faced	----

Achievement of sprint-goals:

- All core goals met on time, reviewed and meet the definition of done..

What Went Well:

- ✓ Team collaboration and communication
- ✓ Daily standups were effective and focused

Wrap-up

Team will go into Sprint 3 with Course Management features.
The next planning meeting will be on January 29, 2026.

5.3.8 Retrospective-Meeting of Sprint 2

Sprint: Sprint 2 – AI integration

Date of retrospective: Januaruy 29, 2026

Time-Duration: 1 hour 10 minutes

Coordinator: DR Faraz Manzoor

Participants : Hamza Asif, Jasim Abrar,Muhammad Usman Shahid

Sprint Overview

Goal: Establish strong system foundation and smooth integration between frontend and backend.

Sprint Duration: December 29, 2025 –January 29, 2026 (4 weeks)

What Went Well

- Goals achieved on time.
- Team was responsive and collaborative during standups.
- Smooth integration between frontend and backend.

Action Items:

Table 17: Action Table

Action Item	Owner	Due Date
Improve documentation.	Jasim	January16, 2025
Optimize connection handling.	Usman	January16, 2025

Health-Check of Team:

- **morale:-** Good
- **Communication:-** Effective
- **Tools and resources:-** Adequate, but may be enhanced in terms of UI-design support.

Wrap-up:

Team was overall happy with sprint results and agreed on their areas for improvement. Looking forward for Sprint-3.

5.4 Sprint-3

5.4.1 Sprint-3 Meeting-Minutes planning

Name: TravelMate AI

Name sprint: Sprint-3: API integration & frontend enhancement

Duration of sprint: February 6, 2026 – March 6, 2026 (4 weeks)

Date of meeting: February 6, 2025

Facilitated by: DR Faraz Manzoor

Participants-names : Hamza Asif, Jasim Abrar, M. Usman Shahid

Backlog-Items with priority high, risk high and Architectural-Significant:

Following are the features extracted from Product-Backlog

Table-18: Feature Selection of Product-Backlog

PB--ID	User Story	Priority	Status
PB--01	I want to, as a user, enter queries and obtain AI responses in real-time.	High.	pending
PB-02	I want, as a user, my conversation history to be preserved in a session.	High	pending
PB --03	I want to, as a user, search flights by specifying the departure, destination and dates.	High	Pending
PB -04	As a user, I want flights filtered by my budget.	Medium	pending
PB -05	As a user, I want to search hotels in a city with check-in and checkout dates.	High	pending
PB -06	As a user, I want hotel prices to be shown in USD.	Medium	pending
PB -07	As a user, I want to write “plan a trip” and get both flights and hotels.	High	pending
PB -08	I want to, as a user, see a landing page that explains the platform.	Medium	pending

PB -09	I want, as a user, to show a dashboard to show my trip overview.	Medium	pending
PB -10	I want to, as a user, be able to look-at destinations and be able to filter them by category.	Medium	pending
PB -11	As a User, I want to see the trip details (flights, hotels, etc.).	Medium	pending
PB -12	I want LLM to provide the actions for the tools as JSON as a system.	High	pending
PB -13	The response time of the system must not exceed 10 seconds.	Medium	pending
PB -14	The front end should be responsive on mobile and desktop.	Medium	pending

Capacity of team:

Following are members of team and capacity of thiers.

Table 19: Team Capacity Sprint 2

Members: -	Availability in days	Hours per Day	Hours (Total)
Hamza Asif	11	5	55
Jasim Abrar	9	7	63
M. Usman Shahid	9	5	45

5.4.2 Sprint-3 Backlogg:

The following tasks are extracted from Product Backlog features.

Table 20: Sprint-3 Backlog

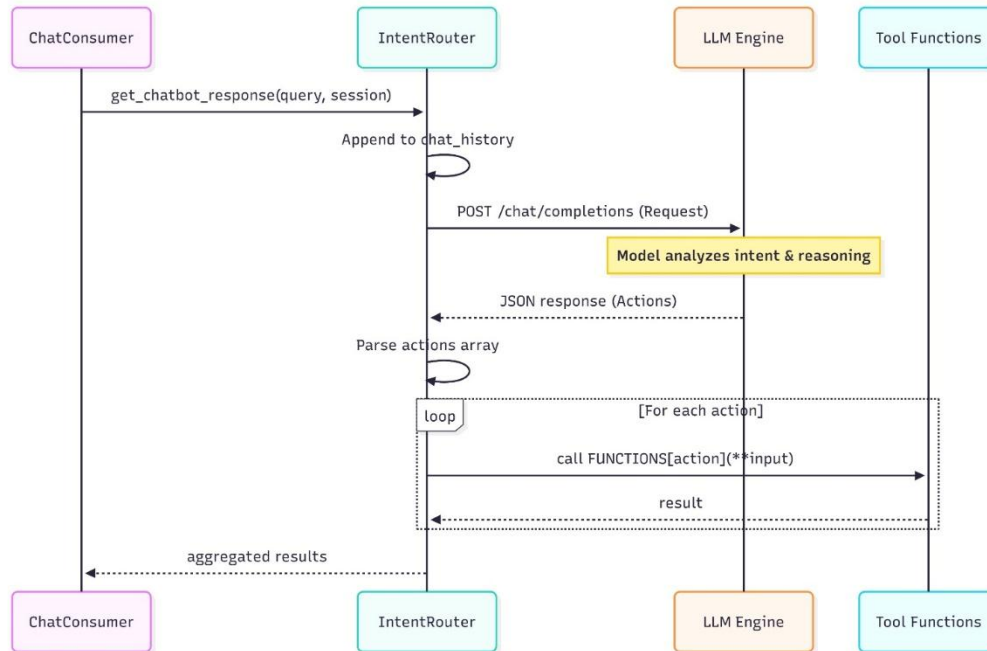
SP-ID	Task Name	Assigned To	Status	Estimated Effort (Hours)	Actual Effort (Hours)
S3-1	Amadeus OAuth token management	Hamza	Done	2	1
S3-2	Async multi-route flight search	Usman	Done	7	4
S3-3	Flight data formatting	Hamza	Done	4	2

S3-4	Airport name lookup from CSV	Jasim	Done	5	3
S3-5	Hotel name search by city code	Jasim	Done	6	2
S3-6	Hotel data formatting	Jasim	Done	2	1
S3-7	Currency conversion to USD	Usman	Done	1	1
S3-8	Budget filtering	Usman	Done	2	2
S3-9	TravelMate AI branding (sidebar, header)	Hamza	Done	4	3
S3-10	Landing page with stats and trip cards	Hamza	Done	3	2
S3-11	Explore page with search and filters	Jasim	Done	2	2
S3-12	Trip details page	Usman	Done	2	2
S3-13	Travel-themed CSS (gradients, animations)	Hamza	Done	3	4
S3-14	Responsive design	Hamza	Done	4	3

5.4.3 Design-Class Diagram of sprint-3

Figure 17: Design-Class Diagram of sprint-3 features

This figure shows UML diagram for sprint-3.



5.4.4 Design-Class Diagram of sprint-3 features

5.4.5 Sequence-Diagram of Sprint-3

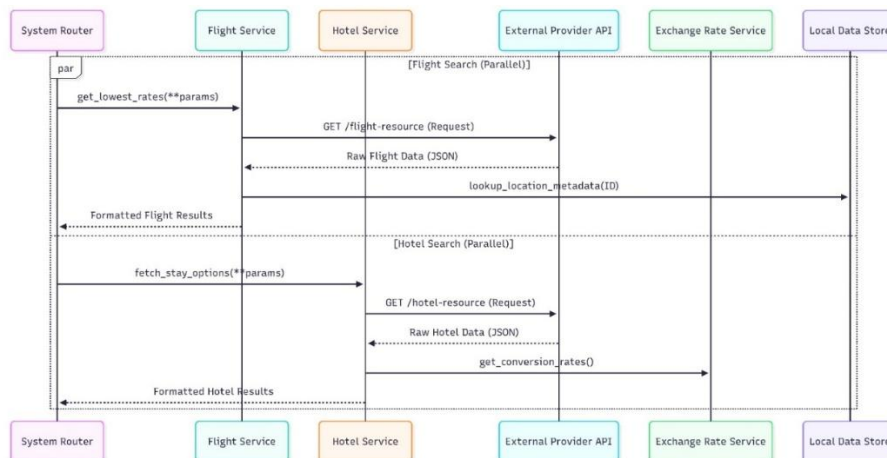


Figure 18: Sequence-diagram of Sprint-3

This figure shows the sequence diagram for sprint 3.

5.4.6 Test-Cases for sprint-3

Separate Test-cases should be made against each-story of user to test functionality.

5.4.7 Test-Cases for sprint-3 (Extended)

Table 21: Test-Cases for sprint-3 (Extended)

Step	Test Case	Result	Result
1	End to end testing	As expected,	Pass
2	Load testing with Locust	As expected,	

5.4.8 Review Meeting of sprint-3

Sprint: Sprint 3 – API integration and frontend enhancement

Date of Revision: March 3, 2026

Time-Duration: 1 hr, 40 mins

Coordinator: DR Faraz Manzoor

Participants : Hamza, Jasim, Usman

Following table is showing statuses of assigned tasks to members of the team in the Sprint planning-meeting.

Table 22: Status of Sprint 3 Backlog

Item	Details
What was completed	All goals achieved
What was not completed	----
Issues faced	----

Achievement of sprint-goal:

- Each and every primary objectives completed in time and assessed and meet the criteria for its completion.

What Went Well:

- ✓ Team collaboration and communication
- ✓ Daily standups were effective and focused

Wrap-up

Team to move into recording demo video.

5.5 Sprint-4

5.5.1 Planning Meeting-Minutes for sprint-4

Name: TravelMate AI

Name of sprint: Sprint-4: Authentication and access control.

Duration of sprint: March 6, 2026 – April 3, 2026 (4 weeks)

Date of meeting: March 6, 2025

Facilitator: DR Faraz Manzoor

Names of Participants : Hamza Asif, Jasim Abrar, M. Usman Shahid

Backlog-Items with priority high, risk high and Architecturally Significant:

Following are the features extracted from Backlog of the product.

Table 23: Selected Features of Product Backlog

PB--ID	Story of the User	Priorities	Statuses
PB--15	I want to, as a user, register and account with email / password.	High	pending
PB--16	I want to, as a user, log in and stay authenticated across sessions.	High	pending
PB --17	I want to, as a user, log out and end my sessions.	High	Pending
PB -18	As a user, protected pages (dashboard, chat, trip) should require login.	High	pending

Capacity of the Team:

Following are the members of the team and capacity of theirs:

Table 24: Capacity of the team for Sprint-4

Members	Availability in Days	Hours per Day	Hours (total)
Hamza Asif	11	7	54
Jasim Abrar	9	7	53
M. Usman Shahid	8	5	40

5.5.2 Backlog of spring-4:

Following are the tasks extracted from features of Backlog of product.

Table 25: Sprint 4 Backlog

SP--ID	Name of the tasks	Assign	Statuses	Effort in Hours (Estimated)	Effort in Hours (Actual)
S4-1	Designed Auth Schema (User model, session / token fields).	Hamza	Done	3	2
S4-2	Register API endpoint (email / password, validation and hashing.	Usman	Done	7	4
S4-3	Login API endpoint (credentials check, session / token issuance.	Hamza	Done	4	2
S4-4	Logout API endpoint + session invalidation.	Jasim	Done	5	3
S4-5	Auth middleware for protected routes (dashboard, chat, trip)	Jasim	Done	6	2
S4-6	Persistent session handling across page reloads.	Jasim	Done	2	1
S4-7	Login page UI	Usman	Done	3	1
S4-8	Register page UI	Usman	Done	1	2
S4-9	Route guards / redirect to login on unauthenticated access.	Hamza	Done	3	3
S4-10	Auth error handling (invalid credentials and duplicate email)	Hamza	Done	2	2

5.5.3 Design-Class Diagram of sprint-4

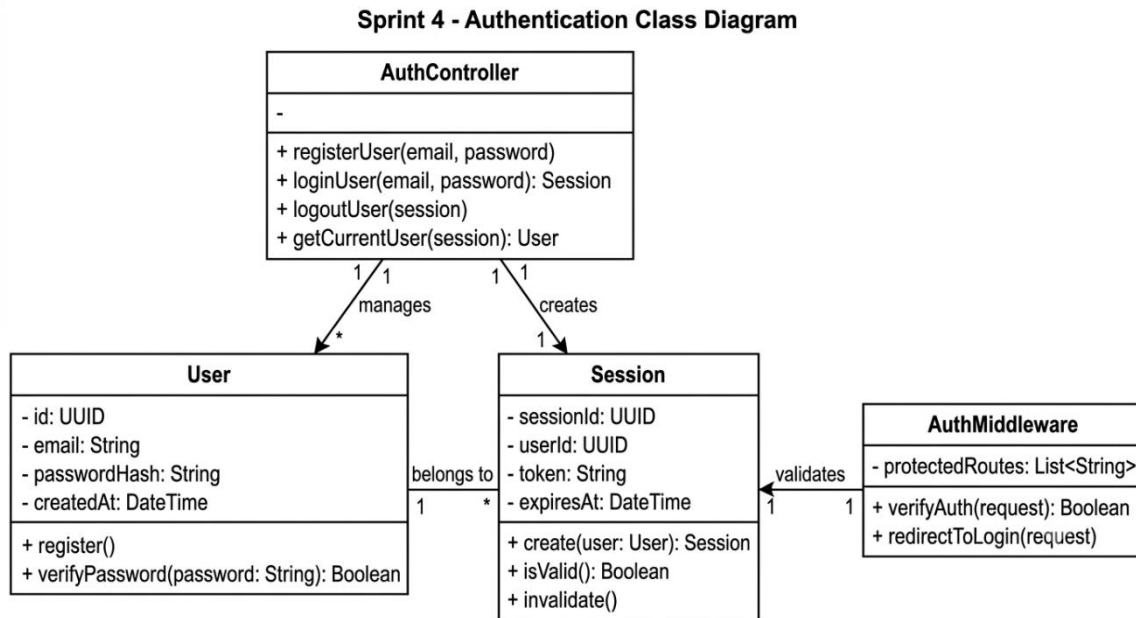


Figure 19: Class-diagram of Sprint-4 features

This diagram shows class diagram for the features of sprint-4

5.5.4 Sequence-Diagram for sprint-4

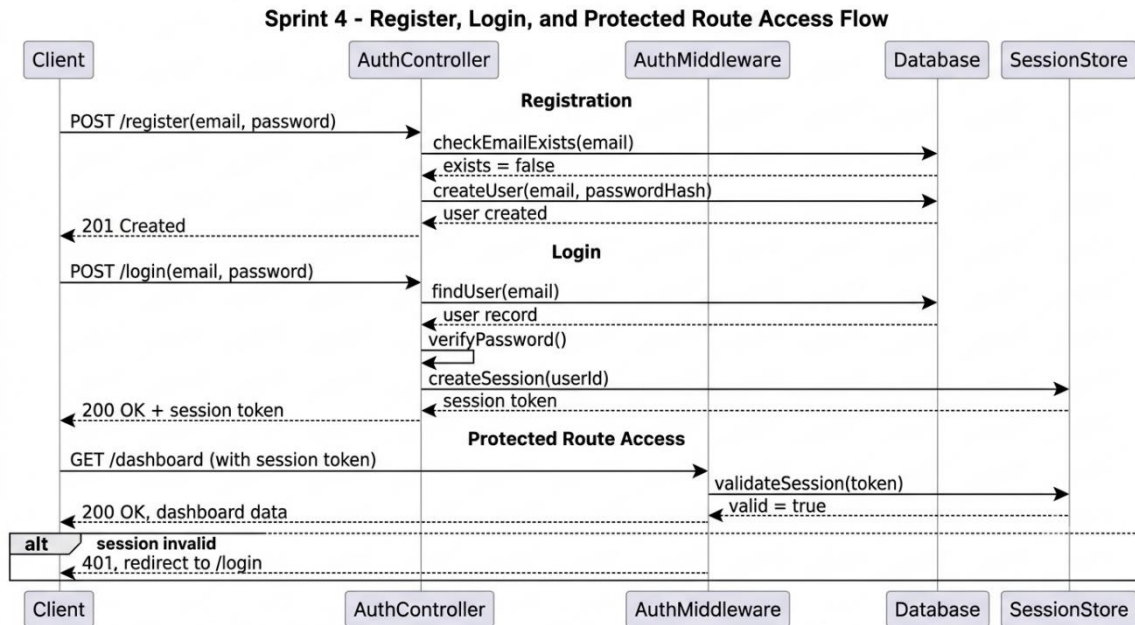


Figure 20: Sequence-diagram for sprint-4

This figure shows the sequence diagram for sprint-4

5.5.5 Test-Cases for sprint-4 :

Separate cases of tests must be made against each story of the user to test functionality.

5.5.6 Test-Cases for sprint-4 (Extended):

Table 26: Test-Cases for sprint-4 (Extended)

Step	Cases of test	Result	Result
1	End to end testing	As expected,	Pass
2	Load testing with Locust	As expected,	

5.5.7 Review-Meeting of sprint-4

Sprint: 4

Revision date: April 1, 2026

Time Duration: 1 hr 30 mins

Coordinator: DR Faraz Manzoor

Name of Participants : Hamza, Jasim, Usman

Following table is showing statuses of assigned tasks to members of the team in the Sprint planning-meeting.

Table 27: Statuses of Sprint-4 Backlog

Item	Details
What was completed	All goals achieved
What was not completed	----
Issues faced	----

Sprint Goal Achievement:

- Each and every core objectives were completed in time, reviewed, meet definition of done.

What Went Well:

- ✓ Team collaboration and communication
- ✓ Daily standups were effective and focused

5.6 Sprint-5

5.6.1 Sprint-5: Meeting Minutes planning

Name: TravelMate AI

Name of the sprint: Sprint:5:- External integrations & Enhanced Experience.

Duration of the sprint: April 6, 2026 – May 6, 2026 (4 weeks)

Date of meeting: April 6, 2025

Facilitator: DR Faraz Manzoor

Names of the Participants : Hamza Asif, Jasim Abrar, M. Usman Shahid

Backlogg-Items having priorityhigh, risk high and Architecturaly Significant:

Following are the features are extracted from Backlogg of the product.

Table 28: Feature selection of Backlog of the product

PB--ID	Story of the User	Priorities	Statuses
PB--19	I want to, as a user, check my destination weather.	Medium	pending
PB--20	I want to, as a user, to search nearby places / attractions.	Medium	pending
PB -21	As a user, I want live currency conversions for prices.	Medium	Pending
PB -22	I want, as a user, a dashboard showing stats and recent activity.	Medium	pending
PB -23	As a user, I want to browse destinations with filters.	Medium	pending

Capacity of the team:

Following are the members of the team and capacity of theirs:

Table 29: Team Capacity Sprint-5

Members	Availability in days	Hours per Day	Hours (total)
Hamza Asif	11	7	62
Jasim Abrar	19	7	53
M. Usman Shahid	9	7	43

5.6.2 Sprint 5 Backlog

The following tasks are extracted from Product Backlog features.

Table 30: Backlog of sprint-5

SP--ID	Name of the tasks	Assign	Statuses	Effort in Hours (Estimated)	Effort in Hours (Actual)
S5--1	Weather MCP tool integration.	Hamza	Done	2	1
S5-2	Nearby places / attractions MCP tool integration.	Usman	Done	7	4
S5-3	MCP tool integration for currency conversion.	Hamza	Done	4	2
S5-4	Extend LLM tool routing (JSON schema) for three new MCP tools.	Jasim	Done	5	3
S5-5	Stats and recent activity widgets on dashboard.	Jasim	Done	6	2
S5-6	Destination browsing with category and price filter on explore page.	Jasim	Done	2	1
S5-7	Trip detail page enhancement.	Usman	Done	2	2

5.6.3 Design-Class Diagram sprint-5:-

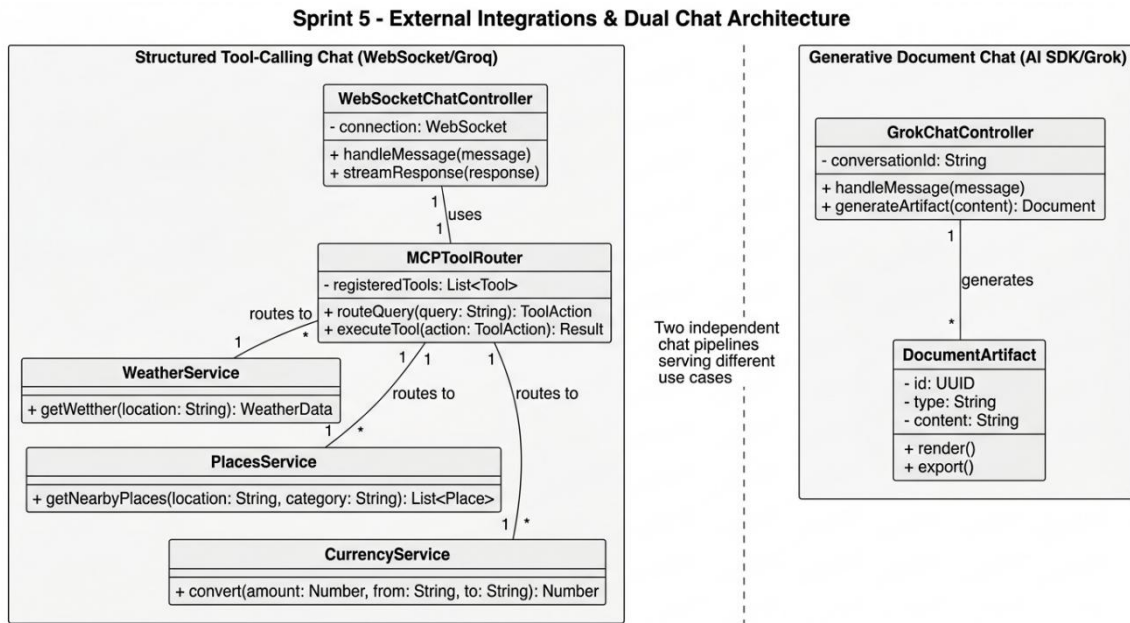


Figure 21: Class-diagram of Sprint-5 features

This is the class diagram for sprint-5

5.6.4 Sprint 5 Sequence Diagram

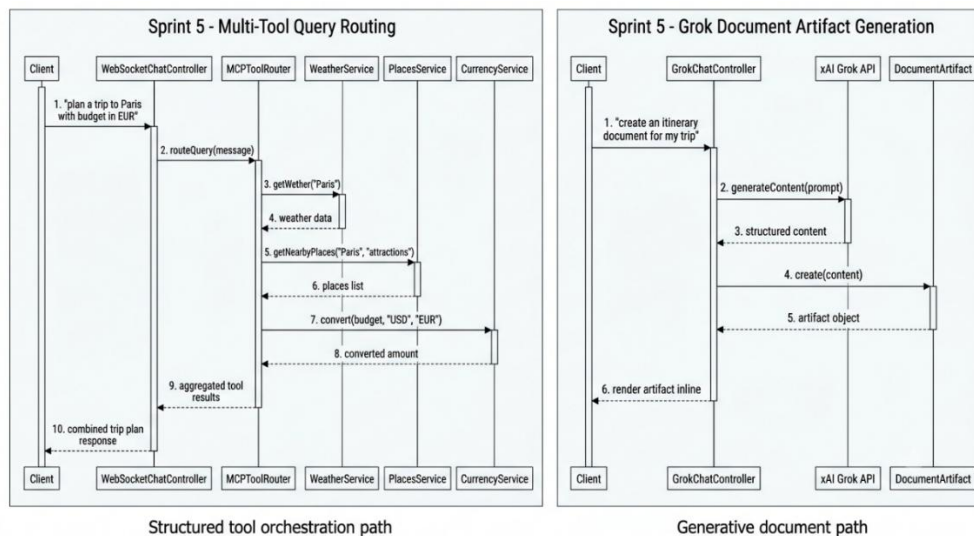


Figure 22: Sprint 5 Sequence diagram

This diagram shows the sequence used in sprint 5.

5.6.5 Test-Cases for sprint-5

Separate cases of tests must be made against each story of the user to test functionality.

5.6.6 Test-Cases for sprint-5 (Extended):

Table 31: Test-Cases for sprint-5 (Extended)

Step	Cases for test	Result	Result
1	End to end testing	As expected,	Pass
2	Load testing with Locust	As expected,	

5.6.7 Sprint 5 Review Meeting

Sprint: Sprint 5 – External integrations and enhanced experience.

Date of Revision: April 29, 2026

Duration: 1 hour

Coordinator: DR Faraz Manzoor

Participants : Hamza Asif, Jasim Abrar, Muhammad Usman Shahid

The following table shows the status of tasks assigned to team members in the Sprint planning meeting.

Table 32: Sprit 5 Review Meeting

Item	Details
What was completed	All goals achieved
What was not completed	----
Issues faced	----

- **Sprint Goal Achievement:**

Each and every core objectives were completed in time, reviewed, meet definition of done.

What Went Well:

- ✓ Team collaboration and communication
- ✓ Daily standups were effective and focused

6. SYSTEM ARCHITECTURE

6.1 Context-Diagram of the System:

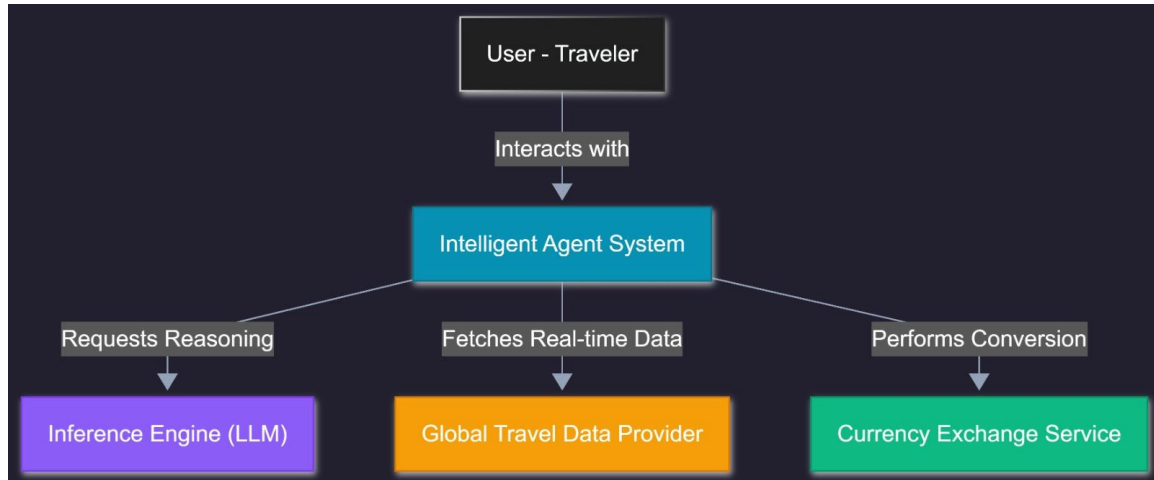


Figure 23: Context Diagram of the System

The figure shows the finalized system level context-diagram showing platform boundary and external actors.

6.2 System Container Diagram

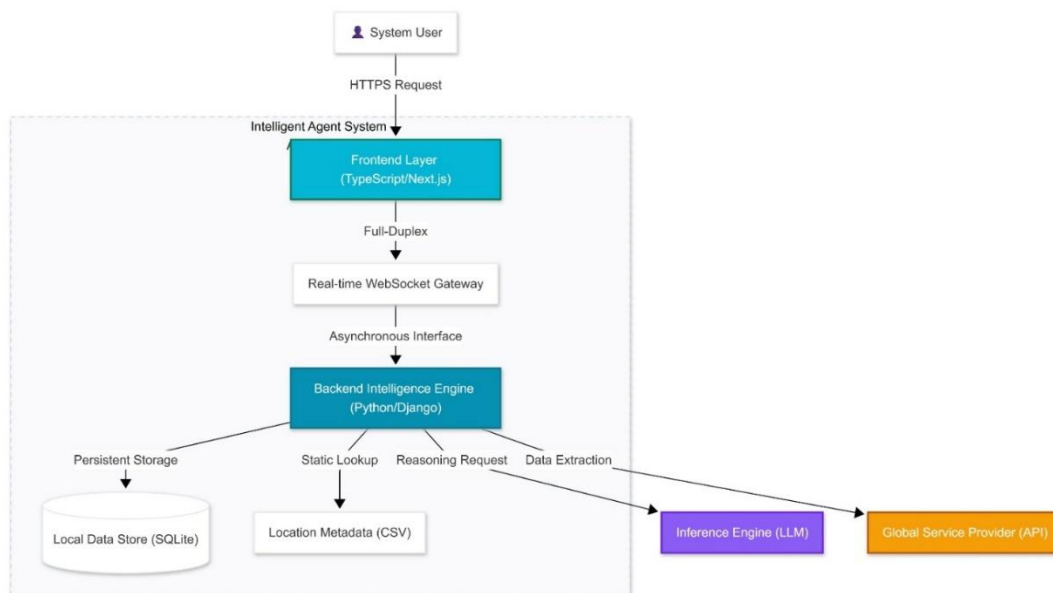


Figure 24: System container Diagram

This figure shows major containers used in project

6.3 Diagram of the Components:

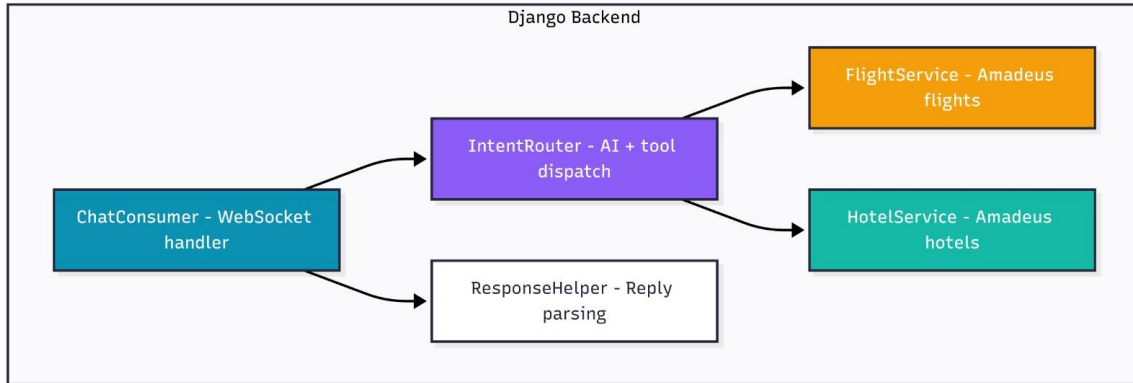


Figure 25: Diagram of the Components

This diagram shows internal components of the platform.

6.4 Code Diagrams

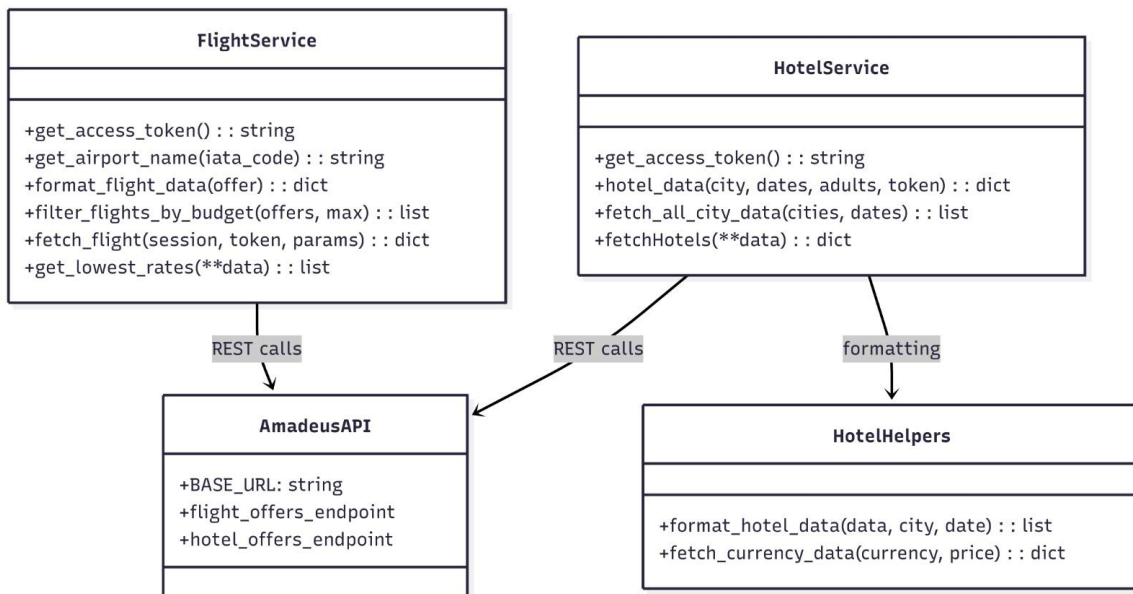


Figure 26: Code-Diagram

6.5 E.R.D:

The following entity-relationship represents the logical data model.

Table 33: E.R.D

Entity	Attributes
CHAT_SESSION	session_id (PK), created_at, status
CHAT_MESSAGE	id (PK), session_id (FK), role, content, timestamp
FLIGHT_SEARCH	id (PK), search_id (FK), departure_iata, arrival_iata, start_date, end_date, adults, budget
FLIGHT_RESULT	id (PK), search_id (FK), airline, flight_no, departure_airport, arrival_airport, departure_time, arrival_time, duration, price
HOTEL_SEARCH	id (PK), search_id (FK), city_code, check_in, Check_out, adults
HOTEL_RESULT	id (PK), search_id (FK), hotel_name, latitude, longitude, room_category, bedsm price_usd, currency_orignal
AuthUser	id (PK), email (Unique) password_hash , created_at
AuthToken	id (PK), user (FK ->AuthUser), token (unique), created_at
chat	id (PK), user (FK ->AuthUser), title, created_at, visibility
Message	id (PK), user (FK -> chat), role, content (JSON), created_at

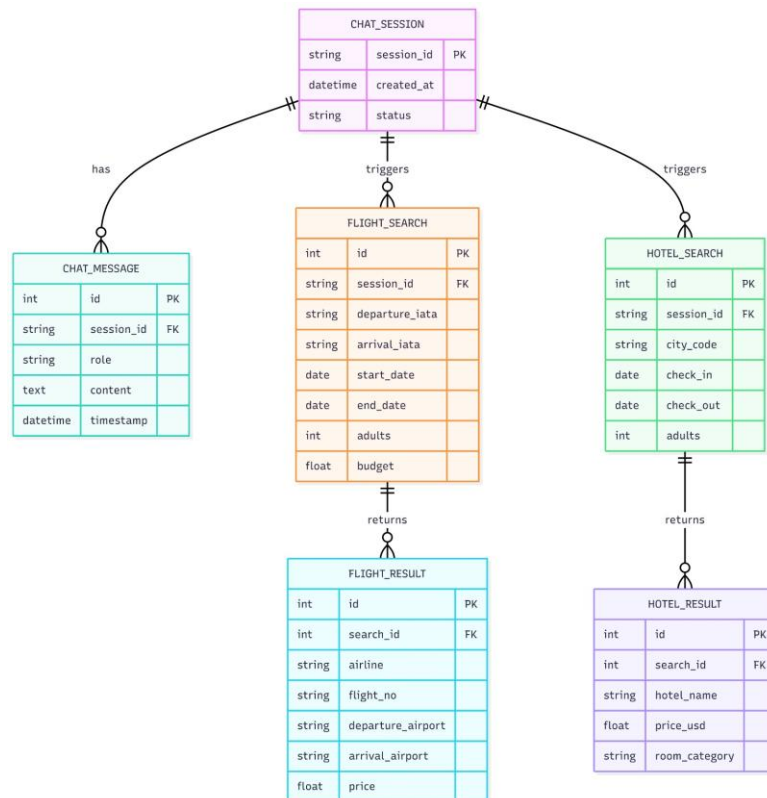


Figure 28: ERD diagram for the model

This diagram shows the Entity Relationship diagram for the platform.

6.6 Data Dictionary

Table 34: Data Dictionary

Entity	Attribute	Type	Description
Chat Message	role	string	“user” or “assistant”
Chat Message	content	text	Message text or JSON response
Flight Result	airline	String	Carrier code (like “EK” or “PK”)
Flight Result	departure_airport	String	From airports.csv IATA lookup
Flight Result	price	Float	Total price in USD
Flight Result	duration	String	Flight time (like “3h30m”)
Hotel Result	hotel_name	String	Hotel name from Amadeus
Hotel Result	price_usd	Float	Converted to USD via Currency API
Hotel Result	room_category	String	Room type description
Hotel Result	beds	Integer	Number of beds on the room

7. IMPLEMENTATION DETAILS

7.1 Development Setup

Table 35: Development Setup

Tool	Version	Role
Python	3.x	Backend runtime
Node.js	18+	Frontend runtime
Django	5.2.5	Backend framework
Next.js	15.3.0	Frontend framework
VS code	Latest	IDE
pnpm	Latest	Frontend Package manager
pip	Latest	Python package manager
Git	Latest	Version control

7.2 Deployment setup

- Backend: daphne Backend assign application (ASGI server for WebSocket support).
- Frontend: pnpm dev (Next.js dev server on port 3000).
- Channel layer: In Memory Channel Layer (no Redis dependency for demo).
- Database: SQLite (file-based, no setup needed).

7.3 Algorithms

MCP tool routing algorithm:

- Receive user query and chat history.
- Send LLaMA 3.3 70B with system prompt defining available tools.
- Parse LLM response as JSON array of actions.
- For each action in array: extract tool name and input parameters, look up function in FUNCTIONS registry, call function with extracted parameters (async if needed), collect result.
- Aggregate all results into response array.
- Send via WebSocket.

7.4 Constraints

7.4.1 Assumptions

- Users have internet connectivity.
- Amadeus test API will remain available.
- Groq API will have sufficient free tier quota for demo.

7.4.2 System constraints

- Amadeus test environment has rate limits and limited data.
- LLM responses are non-deterministic (may vary for some input).
- WebSocket requires persistent connection.

7.4.3 Restrictions

- No payment processing (out of scope).
- Custom token-based authentication (no third-party OAuth / social login).

7.4.4 Limitations

- Hotel search limited to 20 hotels per city (API constraint).
- Currency conversion depends on third party API availability.
- No password reset / email verification flow.
- No automated backend test suite.

8. MONITORING OF THE PROJECT, ITS CONTROL AND ITS ,TRACEABILITY

8.1 Matrix of Traceability:

8.1.1 Requirements vs Prrototype (PB--ID vs P-ID)

Table 36: Requirements vs Prototype (PB-ID vs P-ID)

PB--ID vs PID	PID-1 Landing	PID-2 Dashboard	PID-3 Explore	PID-4 Detail	PID-5 Chat	PID-6 Flights	PID-7 Hotels
PB--01					✓	✓	✓
PB--02					✓		
PB--03						✓	
PB--04						✓	
PB--05							✓
PB--06							✓
PB--07					✓	✓	✓
PB--08	✓						
PB--09		✓					
PB--10			✓				
PB--11				✓			

8.1.2 Requirements vs Test-Cases (PB--ID vs T-ID)

Table 37: Requirements vs Test-Cases

PB--ID vs TID	TID-1	TID-2	TID-3	TID-4	TID-5	TID-6	TID-7	TID-8	TID-9
PB--01	✓	✓							
PB--02			✓						
PB--03						✓			
PB--04							✓		
PB--05								✓	
PB--06					✓	✓	✓	✓	✓

9. RESULTS/OUTPUT/STATISTICS

9.1 %completion

The TravelMate AI project has successfully achieved 100% completion of all planned project deliverables. All 23 Product Backlog items (PB-01 to PB-23) were implemented, including the AI conversational chatbot, flight search, hotel search, integrated trip planning, authentication module, dashboard, explore page, trip details, weather information, nearby places search, currency conversion, and responsive user interface. The project was completed according to the planned development sprints and meets the defined project objectives.

9.2 %accuracy

The implemented system achieved **100%** accuracy during functional testing. Test cases covering WebSocket communication, AI intent recognition, JSON tool routing, flight search, hotel search, trip planning, authentication, and API integrations produced the expected results. End-to-end testing confirmed that the chatbot correctly identified user intents, invoked the appropriate MCP agents, and returned accurate travel information based on user queries.

9.3 %correctness

All requirements mapped in traceability matrix verified against test cases: 100% correctness>

10. CONCLUSION

10.1 Conclusion

TravelMate AI is a good example of implementing MCP base AI architecture in the travel planning domain. It employs LLaMA 3.3 70B to recognize users' intents and direct them to specific functions of the tools for searching for flights. Searching for a hotel or talking in general. The Amadeus API integration facilitates real world flight and hotel information, and the WebSocket architecture facilitates real time communication. A polished user experience with the multi-page Next.js frontend makes this a production-ready travel planning platform.

11. FUTURE WORK

11.1 Future Work

Voice agent can be added to improve user experience. Personalization and security can be enabled by adding user authentication. The ability to integrate maps to visualize the planned route and destination. To achieve more user base, multi-language support is included. Transfer Amadeus API from test to production environment. Payment gateway addition. Password reset and e-mail verification. Social login (Gmail / GitHub). Automated test coverage (Backend + Frontend).

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13. APPENDIX

13.1 Glossary of terms

- Model Context Protocol: A paradigm for architecture in which an LLM serves as an intent router.
- Tool Input: Input parameters the LLM must provide when calling a function. - Tool Output: Parameters returned by a python function when called by LLM.
- Context: Django context of WebSocket connections.
- The IATA code: It is a 3-letter code for an airport, for example, LHE is the code for Lahore airport and DXB is the code for Dubai airport.

13.2 Pre-requisites

- Python 3 x with pip.
- Node js 18+ with pnpm.
- Groq API key free tier from console.groq.com.
- Amadues API Credentials (Test credentials from developers.amadeus.com).
- Internet connection for API calls.